

RequestScheduler

Building
-floors : List<Floor>
-elevatorSystem : ElevatorSystem
+getFloor(floorNumber : int) : Floor

Person
-personId : int
-weight : double
+pressHallButton(Floor : floor, Direction : dir) : void
+pressFloorButton(floor : int) : void

Floor
-floorNumber : int
-upButton : HallButton
-downButton : HallButton
-shaftDoors : List<ShaftDoor>
+pressUpButton() : void
+pressDownButton() : void

ElevatorSystem
-elevators : List<Elevator>
-scheduler : RequestScheduler
-statsManager : StaticsManager
+addRequest(request : ElevatorRequest) : void
+assignRequestToElevator(request : ElevatorRequest) : void
+getStatistics() : StatisticsReport

Customer
-id : int
-name : String

Employee
-id : int
-name String

Visitor
-id : int
-name : String

^a Elevator
-elevatorId : int
-cage : Cage
-servicedFloors : List<Integer>
-currentFloor : int
-currentDirection : Direction
-currentState : ElevatorState
-currentWeight : double
-maxWeight : final double = 4000
-maxCapacity : final int = 20
+moveToFloor(targetFloor : int) : void
+openDoors() : void
+closeDoors() : void
+updateSensorData(floor : int, dir : Direction) : void
+isWeightLimitExceeded() : boolean
+isCapacityExceeded() : boolean
+addPassenger(p : Person) : boolean
+removePassenger(p : Person) : void

Cage
-controlPanel : ControlPanel
-currentPassengers : Lis<Person>
+ringAlarm() : void

ControlPanel

Trip
-person : Person
-elevator : Elevator
-entryFloor : int
-exitFloor : int
-entryTime : Date
-exitTime : Data

ElevatorState
+move(targetFloor : int)
+stop()
+openDoors()

StatisticsManager
-trips : List<Trip>
+recordTripStart(p : Person : e : Elevator, entryFloor : int) : void
+recordTripEnd(p : Person : e : Elevator, entryFloor : int) : void
+generateUsageReport() : StatisticsReport

